

Appendix 1 of Meek et al. “Geography and Admixture Shape the Genome-scale Phylogeny of North American *Delphinium*”

New clade definitions

The following clade definitions synthesize morphological descriptions from the Flora of North America (FNA) and related botanical sources. Each clade is defined using a **minimum-clade** definition (the smallest clade containing the internal specifiers and all descendants). Diagnostic characters summarise the most consistent synapomorphies and habitat preferences, while comparison sections highlight how each clade differs from its congeners. Comments identify notable exclusions. All numbered citations refer to the sources listed in the reference list at the end of the document.

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Clade Glaucaltatum

Definition (minimum-clade): The smallest clade containing *Delphinium glaucum*, *D. exaltatum*, and *D. brachycentrum*.

Etymology: The name combines **glauc-** (from *D. glaucum*) and **-altatum** (from *D. exaltatum*), alluding to the glaucous (bluish) bloom on their stems and the elevated, montane habitats in which these species grow.

Diagnostic characteristics

- **Glabrous or glaucous stems.** The stems of *D. glaucum* and *D. exaltatum* are smooth and often glaucous, lacking the pubescence seen in many other larkspurs[1][2]. *D. brachycentrum* also has a clean stem surface[3].
- **Moderate nectar spurs (≈9–17 mm).** Spurs are of intermediate length: *D. glaucum* has spurs 10–15(–19) mm[1]; *D. exaltatum* 9–12 mm[2]; and *D. brachycentrum* 15–17 mm[3].
- **Intermediate inflorescences (5–90 flowers).** Inflorescences typically bear 8–30 flowers in *D. exaltatum*[2], 5–18 in *D. brachycentrum*[3], and 40–90 in *D. glaucum*[1]—well below the very dense racemes of some southwestern species.
- **Blue to pale-lavender flowers.** Sepal color ranges from bluish-purple (*D. glaucum*)[1] to whitish or pale lavender (*D. exaltatum*)[2]; *D. brachycentrum* has dark-blue flowers with rounded sepals[3].

- **Cool, moist montane distribution.** Members of Glaucaltatum inhabit boreal and montane environments—wet thickets and meadows from Alaska to the northern Rockies for *D. glaucum*[1], rocky slopes in eastern deciduous woods for *D. exaltatum*[2], and alpine tundra slopes for *D. brachycentrum*[3].

Comparison with other clades

- **Versus Novomexirockiensis:** Glaucaltatum species are glabrous or glaucous and bear moderate racemes with 5–90 flowers. In contrast, members of *Novomexirockiensis* are robust, often puberulent plants with 20–120(+) flowers in a raceme[11][13].
- **Versus Columbianum:** Columbianum species are generally shorter (10–100 cm) and have smaller racemes (3–50 flowers) with variable flower colors, including yellow or green; Glaucaltatum lacks yellow or green flowers and has a consistent montane distribution[4][33].
- **Versus Barbescens:** Both clades possess mid-length spurs, but *Barbescens* species often have puberulent inflorescences or leaves and occupy high-elevation wet meadows. Glaucaltatum is characterised by glabrous stems and slightly fewer flowers[17][18].
- **Versus Californicum:** Californicum species inhabit low-elevation chaparral and oak woodlands and have longer nectar spurs (up to 24 mm) and variable sepal colors (lavender, white, pink, red). Glaucaltatum lacks these traits[7][8].
- **Versus Aridimontanum:** Aridimontanum species are short, basal-leaved Great Basin endemics with small racemes and unwinged seeds[20][21]. Glaucaltatum, by contrast, has longer racemes and occurs in moist montane habitats.

Comments

Exclusion: *D. carolinianum* is explicitly excluded from Clade Glaucaltatum. *D. carolinianum* has moderate stems (20–90 cm), variable sepal colors and relatively few flowers (3–27), features that do not match the glabrous, montane phenotype of Glaucaltatum[23].

Summary

Clade Glaucaltatum comprises glabrous or glaucous larkspurs with intermediate-length spurs, moderate racemes, blue to pale-lavender flowers and a preference for cool montane habitats. These traits distinguish it from the taller, densely flowered southern Rocky Mountain clades, the diverse chaparral group and the small Great Basin taxa.

Clade Columbianum

Definition (minimum-clade): The smallest clade containing *Delphinium lineapetalum*, *D. multiplex*, *D. xantholeucum*, *D. nuttallianum*, *D. distichum*, *D. bicolor* and *D. viridescens*.

Etymology: Named for the Columbia Plateau. Most members occur in the mountains surrounding the Columbia Plateau in eastern Washington, north-eastern Oregon, Idaho and western Montana.

Diagnostic characteristics

- **Moderate stature with basal or proximal leaves.** Species range from 10–100 cm tall and typically have leaves restricted to the basal or lower half of the stem. *D. lineapetalum* has glabrous stems 30–60 cm tall with small, deeply lobed leaves[4]; *D. nuttallianum* is 10–40 cm tall with leaves on the proximal quarter[5]; *D. viridescens* and *D. multiplex* are taller (up to 150 cm) but still bear most leaves on the lower half[6][35].
- **Small racemes with 3–50 flowers.** Inflorescences typically contain 3–24 flowers in *D. lineapetalum*[4] and 8–30 in *D. distichum*[31]; even the largest species (*D. viridescens*, *D. multiplex*) rarely exceed 80 flowers[6][35].
- **Moderate spurs (8–23 mm).** Spur length varies across the clade but never reaches the extreme lengths seen in chaparral species: *D. nuttallianum* has spurs 8–23 mm[5]; *D. lineapetalum* 11–17 mm[4]; *D. bicolor* 13–23 mm[32].
- **Diverse flower colors.** Flower color ranges from dark blue (*D. bicolor*, *D. distichum*)[32][31] through lavender (*D. lineapetalum*)[4] to yellowish or greenish (*D. xantholeucum*, *D. viridescens*)[33][6].
- **Habitat affinity for montane forests, meadows and sage-steppe.** Members occur in open pine woods, dry meadows, wet meadows and grassy sage scrub around the Columbia Plateau[4][5][33].

Comparison with other clades

- **Versus Novomexirockiensis:** Columbianum species are shorter and have much smaller racemes (3–50 flowers) than the robust, 20–120+ flowered species of *Novomexirockiensis*[11][13].
- **Versus Californicum:** Californicum species have longer racemes (up to 90 flowers), longer spurs (7–24 mm) and occur in chaparral or oak woodlands at low elevations[7], whereas Columbianum occupies montane or subalpine meadows and includes yellow-flowered taxa.
- **Versus Barbescens:** Barbescens species are generally taller with more robust racemes and uniform blue-purple flowers[17][18]. Columbianum includes low-stature, short-flowered species and a broader color palette.
- **Versus Glaucaltatum:** Columbianum lacks the glabrous, glaucous stems typical of *Glaucaltatum* and includes species with pubescent stems and leaves[4].
- **Versus Aridimontanum:** Aridimontanum is restricted to the Great Basin and has small racemes, moderate spurs and consistently dark-blue flowers[20][21]. Columbianum is more diverse in color and occurs in moister environments.

Comments

This clade currently contains seven species (see definition) and has no known exclusions beyond species placed in other clades.

Summary

Clade Columbianum comprises a heterogeneous set of larkspurs from the mountains encircling the Columbia Plateau. They share moderate stature, small racemes with mid-length spurs, diverse flower colors and adaptations to montane forests, meadows and sage steppe. These characters distinguish them from the robust southern Rocky Mountain clades, the chaparral larkspurs and the Great Basin taxa.

Clade Barbescens

Definition (minimum-clade): The smallest clade containing *Delphinium barbeyi*, *D. glaucescens*, *D. occidentale*, and *D. geyeri*.

Etymology: The name is derived from *D. barbeyi*, a prominent species in the clade, and the Latin participle **-escens** (“becoming”), implying a group of species showing affinity with *barbeyi*.

Diagnostic characteristics

- **Moderate to tall stems with large, lobed leaves.** Stems range from 35–150 cm. *D. barbeyi* produces reniform leaves 4–8 × 7–15 cm[17]; *D. glaucescens* has pentagonal leaves 2–9 × 3–14 cm[18]; *D. occidentale* bears blades 5–15 cm across[27]; and *D. geyeri* has pubescent leaves[19].
- **Racemes with 6–50 flowers.** *D. barbeyi* has 10–50 flowers[17]; *D. glaucescens* 10–25[18]; *D. occidentale* often has >25 flowers[27]; *D. geyeri* 6–30[19].
- **Moderate spurs (9–18 mm).** Spurs range from 10–18 mm in *D. barbeyi*[17], 9–13 mm in *D. glaucescens*[18], 10–15 mm in *D. occidentale*[27], and 11–16 mm in *D. geyeri*[19].
- **Blue to purple flowers with robust spurs.** Flowers are uniformly blue to purple across the clade: dark bluish purple (*D. barbeyi*)[17], dark blue (*D. glaucescens*), bright blue (*D. geyeri*)[19]. The hybrid *D. occidentale* may be paler but still falls within the blue spectrum[27].
- **Montane to subalpine distribution.** Members grow in wet meadows, rocky slopes and streambanks at elevations of ~1 400–4 100 m[17][18][19].

Comparison with other clades

- **Versus Glaucaltatum:** Both clades possess mid-length spurs, but Barbescens species often have puberulent inflorescences or leaves and occupy high-elevation wet meadows, whereas Glaucaltatum species are glabrous and inhabit cooler boreal or eastern montane forests[1][18].
- **Versus Columbianum:** Barbescens species are taller (up to 150 cm) and have more robust racemes, with consistently blue flowers. Columbianum includes shorter species with yellow or green flowers and smaller racemes[4][33].
- **Versus Novomexirockiensis:** Novomexirockiensis species produce very long racemes (20–120+ flowers) and grow predominantly in the southern Rocky

Mountains[11]. Barbescens species have fewer flowers and are centred in the northern and central Rocky Mountains.

- **Versus Californicum:** Californicum species inhabit chaparral slopes at low elevations and may bear red or white flowers; Barbescens lacks red-flowered taxa and occurs at higher elevations.
- **Versus Aridimontanum:** Aridimontanum taxa are Great Basin endemics with small racemes and unwinged seeds[20]; Barbescens species have larger racemes and, in some cases, wing-margined seeds (e.g. *D. barbeyi*)[17].

Comments

This clade comprises four species currently recognised in the FNA, in addition to *D. occidentale*, which has been described as a natural hybrid between *D. barbeyi* and *D. glaucum*. Its inclusion reflects that it does not appear to be more involved in hybridization than many other described taxa based on genomic admixture analyses.

Summary

Clade Barbescens encompasses the mid-elevation, blue-flowered larkspurs of the northern and central Rocky Mountains. They are characterised by moderate to tall stems, large lobed leaves, racemes of 6–50 flowers and mid-length nectar spurs. Their preference for wet meadows and subalpine habitats and their uniform blue flowers separate them from the taller southwestern clades, the varied chaparral taxa and the small Great Basin species.

Clade Novomexirockiensis

Definition (minimum-clade): The smallest clade containing *Delphinium sapellonis*, *D. novomexicanum*, *D. andesicola*, *D. geraniifolium*, *D. scopulorum*, *D. robustum*, *D. ramosum* and *D. alpestre*.

Etymology: A portmanteau of **Novo-mexi-** (“of New Mexico”) and **-rockiensis** (“of the Rockies”), indicating that members are centred in the southern Rocky Mountains and adjacent southwestern uplands.

Diagnostic characteristics and cohesion

- **Robust, erect stems with deeply lobed leaves.** Most species reach 50–200 cm in height and bear large, palmately lobed leaves 5–15 cm wide[11][13]. Even the shorter *D. ramosum* (≈70–100 cm) has pentagonal leaves with 5–21 lobes[15], and *D. alpestre*—though only 5–25 cm tall—shares the same leaf architecture[16].
- **Large, many-flowered racemes.** Inflorescences typically bear 20–90 (sometimes up to 180) flowers in *D. sapellonis*, *D. novomexicanum*, *D. andesicola*, *D. geraniifolium* and *D. robustum*[11][12][14][29][13]. *D. scopulorum* has 10–30 flowers but still produces larger racemes than the Columbianum or Aridimontanum clades[30].
- **Dark blue to purple flowers with moderate spurs (8–15 mm).** Sepals are consistently dark blue or purple across the clade, and nectar spurs range between 8

and 15 mm[11][12]. *D. ramosum* has spurs 9–13 mm ascending ~30° above the horizontal[15].

- **Pubescent or puberulent stems, pedicels and fruits.** Many species exhibit puberulence on the stems or inflorescences (e.g. *D. andesicola*, *D. geraniifolium* and *D. scopulorum*)[14][29][30]. Seed coats often have roughened or wing-margined cells[13].
- **High-elevation montane and subalpine habitats.** Members grow in meadows, riparian woodlands and coniferous forests at elevations of roughly 2 000–3 900 m in New Mexico, Colorado, Utah and Arizona[11][12][13]. *D. alpestre* occurs on exposed alpine talus at very high elevations[16].

Comparison with other clades

- **Versus Glaucaltatum:** Novomexirockiensis species are taller and more robust, often puberulent and possess much longer racemes (20–120+ flowers) than the glabrous, moderate-flowered Glaucaltatum species[1].
- **Versus Columbianum:** Columbianum taxa are shorter, produce smaller racemes (3–50 flowers) and often include yellow or green flowers[4][33]. Novomexirockiensis species lack yellow flowers and are consistently dark blue or purple.
- **Versus Barbescens:** Barbescens species inhabit higher, wetter subalpine habitats and have fewer flowers (6–50)[17], whereas Novomexirockiensis species occur in slightly drier montane meadows and possess longer racemes.
- **Versus Californicum:** Californicum species have longer spurs (up to 24 mm) and grow in low-elevation chaparral[7]. Novomexirockiensis species have moderate spurs, robust racemes and high-elevation distributions.
- **Versus Aridimontanum:** Aridimontanum includes much smaller plants with compact racemes and unwinged seeds, adapted to arid Great Basin habitats[20][21]. Novomexirockiensis species are larger and occupy montane forests and meadows.

Comments

This clade currently includes eight species. No additional exclusions are needed beyond the distinction from other named clades.

Summary

Clade Novomexirockiensis unites the tall, many-flowered larkspurs of the southern Rocky Mountains and adjacent Southwestern uplands. They share robust stems with deeply lobed leaves, dense racemes of dark-blue flowers with moderate nectar spurs and a preference for montane meadows and forests. These features distinguish them from the smaller, yellow-flowered plateau species, the glabrous montane species, the chaparral larkspurs and the arid Great Basin taxa.

Clade Californicum

Definition (minimum-clade): The smallest clade containing *D. cardinale*, *D. californicum*, *D. hesperium*, *D. gypsophilum* and *D. parryi*.

Etymology: Derived from *Delphinium californicum*, the most conspicuous species in the clade. The epithet “californicum” signals that these species are centred in California and occupy chaparral and oak-woodland habitats.

Diagnostic characteristics

- **Moderate to tall stems with few basal leaves.** Plants stand 40–160 cm tall and have leaves mostly confined to the upper portions of the stem. *D. californicum* attains (60–)100–160(–230) cm with leaves restricted to the upper 80 % [10]; *D. gypsophilum* 60–100(–150) cm with glabrous, glaucous stems [8]; *D. cardinale* 50–150(–280) cm with leaves mostly cauline [9]; *D. parryi* (10–)40–80(–110) cm and *D. hesperium* 40–80 cm share similar stature [7].
- **Racemes with many flowers and long nectar spurs.** Inflorescences bear 8–90(–162) flowers. *D. californicum* has 40–90(–162) flowers with spurs 7–11(–14) mm [10]; *D. gypsophilum* 15–30(–64) flowers with spurs 7–15 mm [8]; *D. cardinale* 10–40(–80) flowers with spurs 15–24 mm [9]; *D. hesperium* 15–30(–100) flowers with spurs 9–18 mm [34]; *D. parryi* 8–24(–48) flowers with spurs 9–21(–24) mm [7].
- **Highly variable flower colors and unwinged seeds.** Sepals range from lavender-greenish in *D. californicum* [10], to white–pink in *D. gypsophilum* [8], dark blue in *D. parryi* [7] and *D. hesperium* [34], and bright red in *D. cardinale* [9]. Seeds are typically unwinged or only brick-shaped [8] [7].
- **Low to mid-elevation chaparral and oak-woodland habitats.** These species grow on dry slopes, well-drained hillsides and open woodlands in California and the adjacent Southwest. Elevations range from sea level to ~1 500 m [8] [9].

Comparison with other clades

- **Versus Novomexirockiensis:** Californicum species have longer nectar spurs (often >15 mm) and occur at much lower elevations in chaparral or oak woodland. Novomexirockiensis species have shorter spurs and inhabit high-elevation montane meadows [11].
- **Versus Columbianum:** Columbianum taxa are smaller and produce shorter racemes with moderate spurs; some have yellow or green flowers. Californicum lacks yellow flowers and has larger racemes [4].
- **Versus Barbescens:** Barbescens species are taller and occupy wet subalpine meadows, with uniformly blue flowers [17]; Californicum species occur in drier chaparral and include red and white-flowered taxa.
- **Versus Glaucaltatum:** Glaucaltatum species have glabrous or glaucous stems, moderate racemes and blue to pale flowers [1]; Californicum species may be glabrous or glaucous but possess long spurs and extremely varied sepal colors.

- **Versus Aridimontanum:** Aridimontanum species are low-growing Great Basin taxa with small racemes and moderate spurs[20][21]; Californicum species are taller and concentrated in California chaparral.

Comments

Exclusion: This clade explicitly **excludes** *D. patens*, *D. nudicaule* and *D. trolliifolium*. *D. patens* has reflexed lateral sepals, shorter racemes and ascending spurs[24]; *D. nudicaule* has scarlet tubular flowers with very long spurs and moist cliff habitats[25]; and *D. trolliifolium* has exclusively cauline leaves and elongated fruits in wet oak woods[26]. These traits are absent from Californicum and warrant their placement in separate clades.

Summary

Clade Californicum comprises the chaparral-adapted larkspurs of California and the adjacent Southwest. They are moderate to tall perennials with racemes of many flowers bearing long nectar spurs, highly variable sepal colors and unwinged seeds. Their low-elevation chaparral and oak woodland habitats and long spurs distinguish them from the robust mountain larkspurs of *Novomexirockiensis*, the small plateau species of *Columbianum*, the blue-flowered *Barbescens* taxa and the glaucous montane species of *Glaucaltatum*. The clade explicitly excludes *D. patens*, *D. nudicaule* and *D. trolliifolium*.

Clade Aridimontanum

Definition (minimum-clade): The smallest clade containing *D. scaposum* and *D. andersonii*.

Etymology: From Latin *aridus* (“dry”) and *montanus* (“of the mountains”), reflecting the clade’s distribution in the arid, mountainous landscapes of the Great Basin.

Diagnostic characteristics

- **Moderate stature with basal leaves.** *D. scaposum* has scape-like stems 25–50(–65) cm tall with most leaves at the base[20]; *D. andersonii* has erect stems 30–60(–90) cm with leaves mostly on the lower half[21].
- **Small racemes with 10–25(–40) flowers.** The racemes are cylindric and open, bearing few flowers. Pedicels are 0.6–2.5 cm and glabrous in *D. scaposum*[20] and slightly longer (1–4 cm) and sigmoid in *D. andersonii*[21].
- **Mid-length spurs (12–18 mm) and dark-blue flowers.** Flowers are bright dark blue with spurs 13–18 mm in *D. scaposum*[20] and dark blue with spurs 12–18 mm in *D. andersonii*[21].
- **Elongate fruits and unwinged, brick-shaped seeds.** Fruits measure 12–16 mm in *D. scaposum*[20] and 17–32 mm in *D. andersonii*[21]. Seed coat cells are brick-shaped and smooth.
- **Great Basin habitats.** Both species inhabit juniper woodlands, grasslands and cold desert scrub in the Great Basin at elevations of 1 200–2 700 m[20][21].

Comparison with other clades

- **Versus Novomexirockiensis and Barbescens:** Aridimontanum species are shorter (25–90 cm), have small racemes (10–25 flowers) and unwinged seeds, unlike the tall, many-flowered species of Novomexirockiensis[11] or the wet-meadow species of Barbescens[17].
- **Versus Glaucaltatum:** Glaucaltatum species inhabit cool, moist montane environments and have larger racemes with glabrous stems[1]; Aridimontanum species occupy arid Great Basin habitats and have pubescent or glabrous pedicels and unwinged seeds.
- **Versus Columbianum:** Columbianum includes a broader array of flower colors and habitats; Aridimontanum is restricted to dark-blue flowers and deserts of the Great Basin.
- **Versus Californicum:** Californicum species are taller and have much longer spurs (up to 24 mm) and larger racemes[9]. Aridimontanum species are modest in stature with moderate spurs.

Comments

Exclusion: *D. stachydeum* is explicitly excluded from Clade Aridimontanum. Despite also occurring in the Great Basin, *D. stachydeum* has tall stems (70–150 cm), dense racemes of 30–60(–102) bright-blue flowers, wing-margined seeds and grows in moist Artemisia swales[22].

Summary

Clade Aridimontanum contains the two Great Basin desert larkspurs *D. scaposum* and *D. andersonii*. These species are moderate-sized perennials with basal leaves, small racemes of dark-blue flowers with mid-length spurs, elongate fruits and unwinged seeds. They inhabit arid juniper woodlands and cold desert scrub at moderate elevations. The combination of Great Basin distribution, compact inflorescences and unwinged seeds distinguishes Aridimontanum from all other Delphinium clades described here.

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